

Shift Work: How to Boost Sleep and Health

How Shift Work Harms Sleep & Health: What Helps? ▾

WORK
Harms
Sleep
&

Find out how you can cope better with irregular working hours.

What do policemen, pilots and nurses have in common? They all work odd hours or on shifts. As shift workers, they work outside of the standard daylight hours. Working when you are supposed to sleep disrupts your normal physiology, which can take its toll on your health.

Shift Work and Sleep

Like many of our biological systems, our natural sleep-wake cycle is controlled by the body's internal clock, explains [Dr Licia Tan](#), Associate Consultant, [Department of Occupational Medicine](#), Sengkang General Hospital.

This internal body clock is a repeating 24-hour circadian rhythm, and is influenced by light changes in our environment. When it works right, the circadian rhythm can promote restorative sleep. Shift work regularly disrupts this internal sleep-wake cycle by demanding wakefulness during the night when our internal clock is sending signals to promote sleep.

"As the body's internal clock initiates the production of melatonin, a hormone which promotes sleep, long-term disruptions in the clock may lead to altered melatonin production," clarifies Dr Tan. This can give rise to serious sleep issues, such as having difficulty falling asleep, staying asleep, and causing excessive sleepiness at unwanted times.

Impact On Health

Shift work, particularly rotating shift work, has also been linked to higher risks for serious health problems. Studies suggest that sleep loss or a change in sleeping time pose a metabolic challenge to the body, notes Dr Tan. This may lead to increased blood pressure, high blood sugar, excess body fat around the waist, and abnormal cholesterol or triglyceride levels. This increases the risk of heart disease, stroke, and type 2 diabetes.

Other theories attribute the predisposition to heart disease to the stress of shift work or lifestyle factors, such as the type of food eaten and the time it is eaten. This is worsened when combined with other factors like obesity, smoking and an unhealthy diet.

Many shift workers suffer from gastrointestinal issues like indigestion, gastritis, colitis and peptic ulcer, irregular bowel movements, heartburn, and so forth. These have been attributed to the disruption of the circadian rhythm of the gastric system.

On top of this, female shift workers may experience irregular and painful menstrual cycles, reveals Dr Tan. Those who wish to get pregnant may have delays in conceiving and lower rates of pregnancies, and are at higher risk of spontaneous miscarriage, pre-term delivery, delayed foetal development and babies with low birth weight. Shockingly, women also face a higher risk of breast cancer.

Apart from physical health, shift work may also affect one's psychological well-being. A large research study reported a link between shift work and increased overall risk of adverse mental health outcomes, such as depression. Female shift workers were reported to be most vulnerable.¹

Despite the many health risks, all is not bleak as some individuals have demonstrated reasonable adaptation to shift work. Younger males and those who go to bed later and wake up later seem to be better candidates for shift work.

In addition, it is important to note that shift workers can reduce their health risks by adopting healthy eating habits, engaging in sufficient physical activity, avoiding smoking, and consuming less alcohol.

Tips for Sound & Restful Sleep

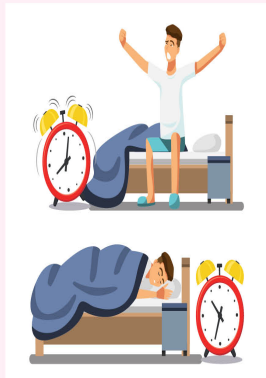
- Night shift workers can reduce their light exposure after leaving work if they are going to sleep once they return home (e.g. they can wear sunglasses or use glasses with blue-light filters).
- Practise good sleep hygiene habits: avoid stimulation from digital devices an hour before going to sleep, avoid caffeine four to five hours before bedtime, minimise disturbances whilst asleep (ie. let housemates know you will be sleeping by switching your phones to silent mode)
- Try to keep to the same sleep-wake schedule on working days and days off. A regular routine helps the body know when to stay alert and when to wind down for sleep.
- Speak to your medical provider about melatonin supplements. They may be effective a few hours before one plans to sleep.

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How Employers Can Help

Generally, clockwise or forward rotating shifts where workers start at a later time than their previous shift are easier to adapt to, as it is easier to stay up late than to go to bed early.

Employers can do their part to promote a conducive work environment for shift workers. They can provide them with healthy food options at all times of the day and night, and opportunities to engage in physical activity and exercise through workplace programmes, recommends Dr Tan.

Taking short naps at the workplace may promote alertness and reduce the risk of work-related injury. If practically feasible, proper scheduling of rest times and appropriate facilities should be provided to maximise worker benefit yet prevent operational disruptions.

Employers should also employ strategies to reduce the mental health burden on shift workers by giving them more autonomy over shift schedules, reducing overall job strain and providing easy access to psychological support services.¹

1. Torquati L, Mielke GI, Brown WJ, Burton NW, Kolbe-Alexander TL.
Shift Work and Poor Mental Health: A Meta-Analysis of
Longitudinal Studies. Am J Public Health. 2019;109(11):e13-e20.
doi:10.2105/AJPH.2019.305278



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